

A Theoretical Model of Hallucinations in NLP Large Language Models

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Abstract

NLP Large Language Models have had a deep impact not only on research but also on society. However, there is one problem underlying their own nature which forbids them to be fully trusted and accountable: Hallucinations – a phenomenon that calls for a scientific account in the form of a theoretical model that represents the underlying rationale of how it emerges. In this paper, we present such a model. In the same spirit as in other fields proposing models of natural phenomena – such as rainfall – or social phenomena – such as knowledge creation in organizations – we propose a theoretical model that serves to explain hallucinations by synthesizing in an abstract and structured framework how the interrelations and interactions of more than 50 factors give rise to this phenomenon helping us to better understand its complexity and idiosyncrasies.

1 Introduction

Large Language Models (LLMs) have become ubiquitous systems across several domains such as Law (Nguyen, 2023), Medicine (Sorin et al., 2023), or Finance (Wu et al., 2023). However, one major obstacle to becoming fully trusted by users is the generation of hallucinations – nonfactual or non-sense outputs (Kaddour et al., 2023; Huang et al., 2025). Hallucinations can have negative impacts on society by, for example, imputing false crimes to people (Lomas, 2025), inventing law cases (Taylor, 2024), or suggesting users to ingest potentially poisonous chemicals (Eichenberger et al., 2025).

Previous works have shed light on possible causes of hallucination such as noisy training data (Ji et al., 2023), stochastic decoding methods (Huang et al., 2025), or LLMs not faithfully learning the data distribution (Lin et al., 2022). In turn, mitigation methods have been proposed such as improved decoding (Shi et al., 2024) and improved training (Lightman et al., 2024) methods.

These efforts have made LLMs less brittle to hallucinations. However, there is a fundamental, open problem: The scarcity of theoretical accounts that substantiate this phenomenon and bring conceptual understanding. Hallucinations from LLMs are not just a simple byproduct of the LLMs’ training data nor do they arise from the effect of isolated components; hallucinations are a complex phenomenon emerging from an intricate, systematic pattern of dependencies and interactions among the LLMs’ underlying components – including the data.

Let us consider two phenomena as illustrative examples. The phenomenon of rain is not independently caused by single factors; it is a set of interrelated factors including evaporation, condensation, temperature, humidity, atmospheric pressure, etc. that make rain to emerge (Pinho and Andrade, 1998; Nugent et al., 2019). In the social sciences, the phenomenon of knowledge creation in organizations is dependent on a framework of interrelated factors across cognitive, relational, and structural dimensions (Nahapiet and Ghoshal, 1998). As Poincaré argued, science not only advances by accumulating facts derived from works studying single factors, but by structuring them into models that allow us to theorize relations and patterns (Poincaré, 1913) – models that can enrich the NLP field to continue its scientific development.

In this paper, we propose a theoretical model of hallucinations in NLP LLMs. Our model aims to conceptually explain the emergence of this phenomenon via an abstract, high-level representation of how factors from LLMs systematically interrelate to cause hallucinations. Besides explanation, our theoretical model can serve multiple roles for future work. First, as a structured framework of previous works providing a visual aid for better informed surveys. Second, as a roadmap for planning both future experiments based on theorized relations and more complex mitigation methods that take into account the dependencies among fac-

tors. Third, as an abstract space where researchers can advance the conceptual knowledge of this phenomenon by discerning and deliberating proposed relations and by further theorizing on new factors and relations, thereby refining our comprehension of this phenomenon.

2 Background and Previous Works

2.1 Hallucinations in LLMs

The phenomenon of hallucinations has been framed as a tripartite concept (Huang et al., 2025; Zhang et al., 2025a; Ji et al., 2023): 1) Nonfactual hallucinations: LLMs’ generations are factually wrong or contradict information in well-established KBs or encyclopedias; 2) unfaithful hallucinations: LLMs’ generations contradict the input prompt or the generated output; and 3) unverifiable hallucinations: LLMs’ generations that cannot be verified due to the lack of reliable sources. We focus on the first two types of hallucinations since we believe are the most relevant in NLP research and applications.

Previous works have studied how candidate factors have played a role in the generation of hallucinations in NLP LLMs, and deep learning models, such as instruction-tuning (Pal et al., 2023), using maximum-likelihood for training (Dziri et al., 2022), or LLMs learning spurious word statistics from pretraining data (Kang and Choi, 2023), among others. However, the literature is lacking a theoretical model connecting these factors, via causal and associative links, into a structured and abstract representation depicting how the phenomenon of hallucinations arises.

Additionally, other works have studied the problem of hallucinations in LLMs from a theoretical standpoint based on either formal analyses of LLMs training process (Kalai et al., 2025), formal linguistic formulations of the generations of LLMs (Bouyamourn, 2023), or statistical models of the risk of hallucinations in LLMs (Gumaan, 2025). Nonetheless, these accounts are different in nature from our theoretical model. They are better categorized as formal mathematical or linguistic models which lie in different layers of abstraction compared to our proposed model: Whereas our model uses natural language and a visual aid as the main representational devices for encompassing the structure and logic of the phenomenon of hallucinations, those previous works use theorems and proofs as their main devices to establish necessary conditions for modeling hallucinations.

2.2 What is a Theoretical Model?

A theoretical model is an abstract, high-level, and conceptual representation of a set of systematically interrelated factors or concepts via causal (and other types of) links, which may or may not be grounded on empirical evidence, that depicts how a phenomenon emerges (Saunders et al., 2024) (Saunders et al., 2015). This class of model qualifies as a type of scientific explanation due to being a conceptual and general explanation of a phenomenon (Bhattacharjee, 2019). Although a theoretical model due to its conceptual nature is inherently theoretical, the theorized relationships among factors offer opportunities for empirical validation in future works (Swanson and Chermack, 2013).

3 Modeling Basis and Method for Building Our Theoretical Model

3.1 Choice of Theoretical Model

We use a variant of the theoretical model proposed by (McCain, 2022; Swanson and Chermack, 2013) because 1) it depicts dependencies and interactions between factors; 2) natural language can be used as the representational device for describing the relations underlying the model, as is common in the literature (N. and Rajendran, 2015; Eisend and Kuss, 2019); and 3) it explains how factors can be a direct/indirect cause/influence of the dependent variable – the generation of a hallucination by an LLM; i.e. it shows how factors are responsible for LLMs hallucinations (Figure 1 and Section 4).

3.2 Our Adapted Method for Building our Theoretical Model

Building a theoretical model comprises a subset of the steps for building scientific theories (N. and Rajendran, 2015). Hence, we borrow established theory-building methods from the Social Sciences due to their mature stage after decades of research (Bhattacharjee, 2019; Bacharach, 1989; Eisend and Kuss, 2019). Specifically, we adapt the deductive methods of (Swanson and Chermack, 2013; Whetten, 1989; N. and Rajendran, 2015), as well as the inductive methods of (Meza et al., 2023; Eisenhardt and Graebner, 2007) since it is difficult that a single method fits all possible scenarios for building a model and so methodological tailoring is required (Eisenhardt, 1989; Swanson and Chermack, 2013).

More concretely, we first review the NLP literature of LLM hallucinations; then, we select factors which are plausible causes of hallucination while

theorizing or extracting dependencies among them based on both the empirical evidence shown in the respective works and our domain knowledge. This bottom-up strategy allows us not only to synthesize in a structured way previous works, but also to find generalizable patterns while reconciling any possible inconsistency by crossing works and validating the effect of each factor on the generation of hallucinations. Moreover, this strategy also facilitates the extraction or proposal of explanations of why each factor causes the target phenomenon (Eisenhardt, 1989). This results in an emergent model grounded on previous works revealing causal (and other) patterns. After that, we reflect on the rationale underlying the proposed conceptual structure to deduce what possible factors are missing in order to have a coherent and logical model that accounts for a viable explanation of how hallucinations are generated. We then proceed to search for these factors in the literature and add them to our model following the procedure described above. If there are factors for which we were not able to find any previous evidence of their role in causing hallucinations, then we theorize, factor by factor, how they are related to the existing model built so far; this theorizing comes from domain knowledge and from efforts to keep our model both congruent with the existing dependencies and parsimonious. We then revisit our full model to search for possible missing links and to double-check the validity of each connection, as advised by (Rivard, 2021). We thus obtain our model of hallucinations (Figure 1).

3.3 Assumptions and Boundaries

We provide boundaries and assumptions under which our model is aimed to work. Our model is aimed for decoder-only Transformer NLP LLMs pretrained in English language (or multilingual where English is one of the main languages). We only consider the standard training process of LLMs: Pretraining, SFT, human-alignment (further assumptions in Section A.1).

3.4 Types of Relationships

Relations between factors depend on the domain (Salmons, 2019). We identify 6 types of relations: 1) *causal*, which corresponds to relations where the antecedent causes the consequent; 2) *influence* or *associative*, where the antecedent has an influence on the consequent, i.e. the consequent’s state value depends on the antecedent, but the antecedent does not cause the consequent; 3) *embodiment*, which

means that the antecedent’s effect will be embodied by the consequent; 4) *instantiation*, where the consequent is a specific type of factor of the antecedent; 5) *mediation*, where a factor is an intermediary between its antecedent and its consequent (e.g. B in $A \rightarrow B \rightarrow C$) and the effect of its antecedent depends on the state of the current factor; and 6) *interaction*, where two factors mutually modify the strength of their effects making them weaker or stronger depending on their state values. These relationships are differentiated in Figure 1 by color.

4 Theoretical Model of Hallucinations: Explaining Hallucinations

The phenomenon of hallucinations in NLP LLMs can be explained from factors from pretraining, Supervised Fine-tuning (SFT), human alignment training, the architecture of the LLM, and those arising at inference time. Figure 1 depicts our model connecting all those factors showing that some can be direct or indirect causes. We elaborate on this model by accounting for the logic behind what we consider the main candidate causes. In Section A.2, Tables 3 to 15 show all relations in Figure 1, and Tables 1 and 2 show examples of hallucinations, or scenarios where hallucinations can arise, for selected factors.

4.1 Causes of Hallucination from the Pretraining Stage

4.1.1 The Influence of Pretraining Data on Hallucinations

According to the type of **domain of pretraining data**, the states of factors such as **data quality**, **subjective information**, and **amount of pretraining data** may induce hallucinations (Ji et al., 2023; Huang et al., 2025; Li et al., 2024b). For example, social media documents, compared to scientific documents, might be of questionable quality due to containing biases, inaccurate info, noise, low quality of writing (syntactically and semantically), but they are abundant on the web. Thus, the way how the choice of domain data may cause hallucinations is via these features affecting the **parameters** state of an LLM: When trained on social media data, the embedding representations of entities and concepts will not be accurate nor rich in concise information; parameters at higher layers of the LLM may not encode proper grammar rules nor meaningful context information; thus, when these representations are used at inference time with an input prompt contain-

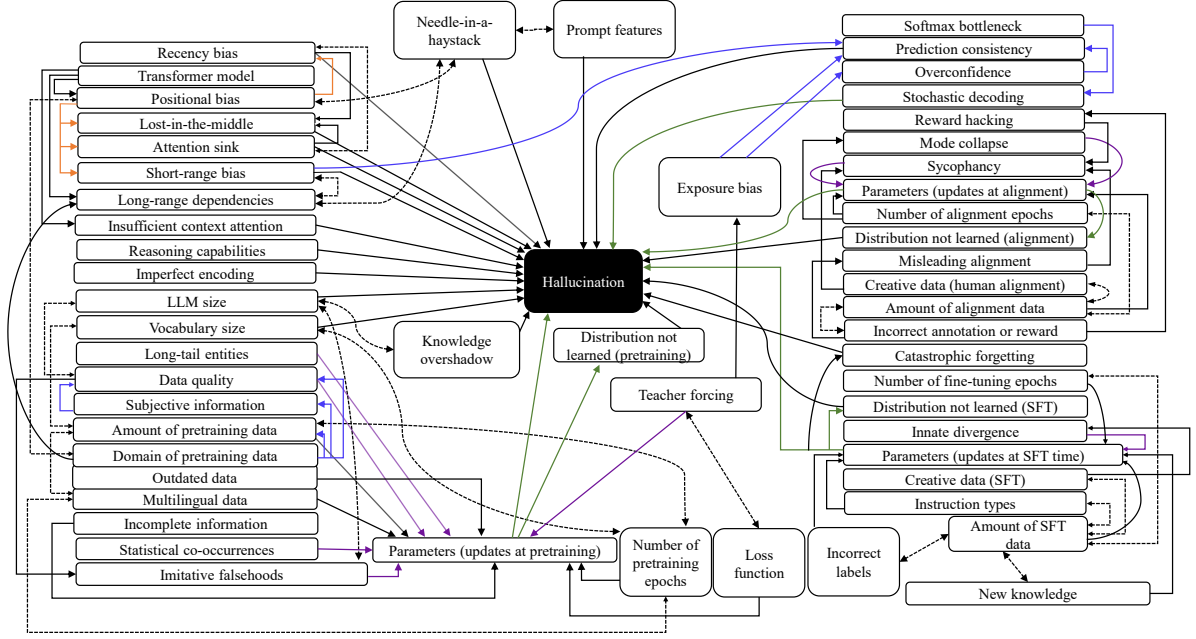


Figure 1: Our theoretical model of hallucinations in NLP LLMs. Single-head arrows indicate these relations: causal (black), influence (blue), embodiment (purple), instantiation (orange), or mediation (green). Double-head, dotted arrows indicate interaction. Description of all relations can be found in Tables 3 to 15 in Section A.2.

ing a novel context accompanying a target entity, the LLM will not be able to accurately extract information, or pay the correct attention weight to each token, leading to a nonfactual hallucination.

Hallucinations are also elicited by other data features. For instance, **outdated data** will allow outdated context information to be encoded in the entities embeddings, leading to a hallucination whenever an LLM provides this outdated information (Huang et al., 2025). Also, data containing **long-tail entities** will result in *poor* embeddings with insufficient context information, resulting in LLMs hallucinating due to the lack of knowledge (Xu et al., 2025). Moreover, **multilingual data** has been shown to impair high-resource language LLMs consequently leading to hallucinations: The variations of linguistic patterns of other languages may increasingly conflict with English patterns the higher the **amount of pretraining data** is used (Chang et al., 2024); i.e. the amount of pretraining data interacts with the multilinguality of the data (scenario in Table 1). Another possible cause of hallucination is the **incomplete information** of entities (Yang et al., 2023; Banerjee et al., 2024), i.e. when no document provides information about a specific aspect of an entity. Thus, when an LLM is queried about a missing aspect, it may respond

based on similar entities resulting in a nonfactual hallucination if, for example, the information requested is the birthday of a person. An additional cause is the frequent co-occurrence of two entities in the corpus – so-called **statistical co-occurrence** (Kang and Choi, 2023). The LLM may learn that regardless of their context, these entities should appear together; e.g., when queried with "What is the capital of Canada?" an LLM may incorrectly reply "Toronto" if Canada and Toronto appeared so frequently together overriding the correct association of capital_of(Canada, Ottawa) in the middle and late-layer self-attention heads (Yu et al., 2024).

4.1.2 The Influence of Pretraining on Hallucinations

Four main causes of hallucination arise as part of, or as a consequence of, pretraining: **Imitative falsehoods**, **teacher forcing**, **loss function**, and the **distribution of the data not being learned**. Inaccurate, noisy information and biases, such as social stereotypes, can be categorized as *falsehoods*, i.e. statements which truth value may not be bounded by reality; pretraining using data with falsehoods may lead to the phenomenon of **imitative falsehoods** (Lin et al., 2022) where an LLM not only memorizes falsehoods but it learns a latent model to mimic them, and by doing so it will halluci-

nate whenever given as input entities associated to such falsehoods. On the other hand, during pre-training, to predict the next token in a sequence, an LLM always receives the golden previous sequence; this **teacher-forced** expectation to receive the correct input sequence will forbid the LLM to learn to recover from wrong predictions; thus, at inference time, incorrect predictions may accumulate deriving into hallucinations (Dziri et al., 2022). Alongside this, the **loss function** (or pre-training objective) qualifies as a plausible factor inducing hallucinations (McKenna et al., 2023). The autoregressive nature of learning incentivizes the prediction of tokens based on statistical and linguistic patterns from the data; it does not enforce factuality (scenario in Table 1).

After pretraining, another cause may arise – indirectly caused by the **number of pretraining epochs** – namely, that the **pretraining distribution was not learned** or was not accurately encoded (Lin et al., 2022); this is a multivariate distribution encompassing variables such as statistics, attributes, and relations of entities and concepts, linguistic and semantic patterns, etc. Thus, not thoroughly capturing this distribution may translate into hallucinations due to insufficient context information or inaccurate patterns learned, such as an LLM not accurately learning the structure and semantics of the perfect present tense possibly due to insufficient exposure (example in Table 1).

4.2 Causes of Hallucination from the Supervised Fine-tuning Stage

4.2.1 The Role of SFT Data on Hallucinations

Similar to the pretraining scenario, some features from the SFT data can induce hallucinations in LLMs: Instances with **incorrect labels** can lead the training of an LLM to a **parameter** space where inputs are systematically mapped to wrong labels (Ji et al., 2023), leading the LLM to hallucinate by generating incorrect outputs as in the case of wrongly translated English-French sentences. Moreover, if the data contains **new knowledge** (Gekhman et al., 2024) unseen during pretraining, to accurately learn it, the training process may significantly alter the pretraining distribution encoded in the **parameters**, leading to **catastrophic forgetting** (Li et al., 2024a) which may cause hallucinations by altering or forgetting internalized knowledge (Zhai et al., 2023).

Moreover, another candidate cause of halluci-

nations is **creative data** (Dziri et al., 2022), such as open-ended dialogues, which enforce the phenomenon of **innate divergence** in LLMs by teaching them to be creative (we elaborate in Section 4.2.2). Finally, if the **amount of data** available for SFT is scarce it will not allow an LLM to accurately picking all the right statistical cues due to an insufficient backpropagation signal to fix errors and learn from them – **parameters** will not be accurately adjusted to learn the tasks correctly – specially for **instruction types** from complex tasks such as multi-hop QA or from different unrelated tasks (Li et al., 2024b); but even for simple tasks, insufficient data can lead to hallucinations (Kang et al., 2025) (example in Table 1).

4.2.2 The Role of SFT on Hallucinations

One of the contributing causes of hallucination from the SFT stage is the phenomenon of **innate divergence** (Ji et al., 2023), generated by dialogue data (and the like), which encapsulates the LLMs’ behavior of inherently diverging from the input context for the sake of creativity. This behavior may result in nonfactual hallucinations when dialogues are grounded in an ill-defined domain with low-quality data, as in the case of recreational data for chatbots where responses are expected to be not only creative but also *funny* and *entertaining* which may contain factually inaccurate statements. Another relevant cause of hallucination is the insufficient (or excessive) **number of fine-tuning epochs** which indirectly contributes to the **SFT distribution not being properly learned** (Pal et al., 2023) by leading to a **parameter** space where the LLM fails to pick necessary statistical, linguistic, and semantic patterns underfitting (or overfitting) the data which may lead to hallucinations (example in Table 2).

4.3 Causes of Hallucination from the Human Alignment Stage

4.3.1 The Impact of Alignment Data on Hallucinations

Features from the data for aligning LLMs to human preferences, similar to the SFT or pretraining stages, can also elicit hallucinations: **Incorrect annotations or rewards**, for DPO (Rafailov et al., 2023) or RLHF (Ouyang et al., 2022) methods, may lead the aligned LLM to prioritize certain aspects – such as fluency, avoiding biases, or being polite – over factuality (Chaduvula et al., 2026). The systematic application of these annotations (or

rewards) will lead to an underlying latent model of the data (or reward) known as **misleading alignment** (Zhang et al., 2025a) (when using DPO) or **reward hacking** (Wang et al., 2025) (when using RLHF), which in turn promotes the internalization of the phenomenon of **sycophancy** (explained in Section 4.3.2). Additionally, another reinforcing factor of sycophancy are **creative data** (Dziri et al., 2022) with an effect similar to that from the SFT stage, which we believe rather than be embodied as the phenomenon of innate divergence, it will be subsumed by the sycophancy phenomenon itself since the goal is to align to human criteria. Lastly, a similar effect from the **amount of data** available for alignment takes place as in previous stages.

4.3.2 The Impact of Alignment to Human Preferences on Hallucinations

Training an LLM to align to human preferences may allow certain factors to cause hallucinations. First, the **sycophancy** phenomenon (Wang et al., 2025) will form a dual internalization of a *servile* behavior possibly at the cost of factuality: Pleasing human values and generating creative outputs. This may give rise to a hallucination whenever an input prompt touches a contentious aspect such as: "Why men seem to hold more job positions in engineering than women?" where a possible response from an aligned LLM might be "that is an interesting but biased insight" when, in fact, the LLM could generate a neutral research-oriented and evidence-based factual response learned from reliable sources.

On the other hand, while an insufficient **number of alignment epochs** may contribute to **not learning the data distribution**, overtraining will lead the **parameters** to encode noise from the data, thus embodying the phenomenon of **mode collapse** (Hamilton, 2024), leading to hallucinations by not learning to generalize to different scenarios. For example, given prompts for alignment from educational scenarios such as "You are an empathetic LLM. Respond to the query while guiding through each step: how to prove the Pythagoras theorem?" the LLM may learn to associate being "empathetic" with educational contexts; thus, at inference time when the LLM receives an input with a medical scenario such as "You are an empathetic LLM. Respond to the query: I have these symptoms... How can I know if this is cancer?" the LLM may hallucinate procedural responses of medical with those of educational contexts and reply "Step 1, can you recognize the variables of the theorem?".

4.4 Causes of Hallucination from the LLM Architecture

Factors in the **Transformer model** such as the attention mask, context window size, the positional encoding, as well as issues such as attention dilution or vanishing gradients contribute to some types of **positional bias** (Wu et al., 2025b), such as **lost-in-the-middle**, **attention sink**, **short-range bias**, or **recency bias**, and to the issues of **long-range dependencies** and **insufficient context attention**; these factors contribute to the generation of hallucinations. **Lost-in-the-middle** (Liu et al., 2024) is the phenomenon where tokens placed in the middle of a long input context do not receive as big attention weights as tokens placed at the beginning or end of the input; this can lead to unfaithful hallucinations, for example, in lengthy input documents if the answer of a given query is placed in the middle. One possible reason for this phenomenon to arise is the presence of **recency bias** (Zhou et al., 2024) which occurs when an LLM puts more attention on recent tokens in the input sequence rather than on prior tokens; this bias may also lead to hallucinations if, for example, the LLM is tasked with summarizing a document and at some point the most recent tokens in the input sequence are those generated by the LLM in which case the LLM may put more focus on such generations. Another possible contributing factor to the lost-in-the-middle phenomenon is the phenomenon of **attention sink** (Gu et al., 2025; Xiao et al., 2024) which happens when some tokens at the beginning of an input prompt receive most of the attention as if they were *pulling* the attention towards them, possibly causing hallucinations when key information required to answer a query is placed towards the end of the input sequence. On the other hand, the **short-range bias** (Kazemnejad et al., 2023) happens due to the similar positional encoding vectors of proximal tokens, which affects their alignment and attention scores influencing them to be higher, leading to bigger attention weights for localized clusters of tokens and to similar hallucinations as the other positional biases. Additionally, an indirect cause of hallucinations is the factor of **long-range dependencies** (Assogba and Ren, 2025; Wu et al., 2025a) where two semantically related but distant tokens share a small, rather than a big, attention weight due to their distance. This may lead to hallucinations in long input sequences by making the LLM to focus on specific localized clusters of the sequence

whereas attention should be put across the whole sequence including both distant tokens which will result in the LLM not properly contextualizing the vectors.

Moreover, it has been shown that some LLMs pay more attention to their parametric memory than to the input context when answering a query or summarizing a document – a phenomenon known as **insufficient context attention** (Shi et al., 2024); this can effectively lead to a hallucination if the correct response to the query or the information required for the summary is contained only in the input context and any information retrieved from parametric memory may contradict it.

A further consideration is any limitation in the **reasoning capabilities** of LLMs which may also induce hallucinations (Zhang et al., 2025a). For example, failure to carry out a complex reasoning throughout input documents to answer complex historiographical queries can lead to hallucinations (example in Table 2). Another contributing factor to the generation of hallucinations is the **LLM size** (number of parameters) (Huang et al., 2025): Large LLMs may generate less hallucinations than small LLMs because the former may have a better capacity for faithfully learning the data distribution.

Additionally, we identify two factors that to our best knowledge have remained unexamined in the LLM literature. Similar to encoding issues in previous NLG models (Ji et al., 2023), the **imperfect encoding** of contextual information in the embeddings of entities and concepts may cause hallucinations; this information is not copy-pasted from the documents, it is constructed and embedded which can lead to loss or deformation of information at any step during the mapping process; this can lead to an incomplete internalization of an entity – not all its features and relations are included in its embedding. Thus, when an LLM is queried about such missing information of an entity it will hallucinate it by responding with information from a similar entity. The second factor we consider is the **vocabulary size** of an LLM. We contend that the finer-grained the vocabulary of an LLM,¹ the less hallucinations: In an ideal scenario each concept has its own embedding which accurately and solely captures its intended contextual meaning, provided that enough **amount of pretraining data** is available; in a less ideal scenario where concepts share

token-level embeddings, the overlap in contexts may lead to ambiguity and in turn to hallucination (example in Table 2).

4.5 Causes of Hallucination Due to Factors at Inference Time

Another set of critical factors contributing to the generation of hallucinations are those arising at inference time. We start our account with the phenomenon of **softmax bottleneck** (Chang and McCallum, 2022) where inside the set of k highest-probability candidate tokens for a prediction, some tokens will be semantically related among them but they may be unrelated to the input query. As demonstrated and analyzed in (Chang and McCallum, 2022), given an input query such as “When the king met the woman, protocol dictated that the [MASK] should be the one to greet first”, any of the two tokens ‘woman’ or ‘king’ would be a faithful choice; nevertheless, due to the softmax bottleneck, a plausible set of $k = 4$ candidate tokens would be ‘king’, ‘queen’, ‘woman’, ‘man’ which is generated due to the geometrical configuration $queen - king = woman - man$. Connected to this phenomenon, the effect of **stochastic decoding** can contribute to a faithfulness hallucination (Dziri et al., 2022) by choosing either the token of ‘queen’ or ‘man’ since methods such as top-k or top-p add randomness to avoid the *likelihood trap*² leading to a more human-like, creative output.

Another cause of hallucination is the phenomenon we term **prediction consistency**, also known as **hallucination snowballing** (Zhang et al., 2024), where LLMs incorrectly predict the first tokens in a sequence generation and rather than predicting subsequent tokens that correct the previous tokens, the LLMs seem to *prefer* to keep consistency in the generation continuing the hallucination (example in Table 2). Aligned to this, LLMs have been shown to exhibit some predictions with **overconfidence** (Zhang et al., 2025a), in the form of high likelihood or probability, despite being hallucinations. This factor may play a role in the phenomenon of **prediction consistency** by reinforcing the hallucinated sequence. Overconfidence when predicting a hallucination may be caused by a latent pattern learned at training time; for example, given the query “in the USA robots can already operate [MASK]”, LLMs may respond “patients” where the correct answer is “vehicles”;

¹The more the number of meaningful entries in the vocabulary added prior to pretraining.

²High-likelihood sequences are more prone to lead to lower-quality text.

this may occur not because of simple contextual features in pretraining data, but because at a higher semantic level the LLM built a representation of robots having near-human capabilities, as reported in some publications and news, leading to confidently predict "patients" to fit in the discourse level the narrative that robots have such advanced abilities. Equally significant is the phenomenon of **exposure bias** (Ji et al., 2023): Due to the mechanism of **teacher forcing** (Section 4.1.2), LLMs were trained by always receiving a gold input sequence for generating the next token; at inference time, the interruption of this scheme may lead to an unpredictable behavior that can influence the quality of the generations. We believe this breach between training and inference could influence the factor of **prediction consistency**: The LLM's divergence from predicting a factual sequence may be stimulated or exacerbated by the **exposure bias** effect by not being able to control or correct the generation due to not having been trained to do so.

In addition, **prompt features** also play a role on the hallucinations of LLMs. Specifically, the **prompt length** can increase the risk of generating hallucinations (Liu et al., 2024; Rawte et al., 2024): Prompts reaching the maximum input length permissible can lead to the problems of long-range dependencies or lost-in-the-middle where key contextual information may be neglected to some extent, forbidding a fully-grounded answer generation, as explained in Section 4.4. Furthermore, **query specificity** or **context precision** can lead to hallucinations as well (Janiak et al., 2025): Vague queries or incomplete contexts can lead to inaccurate answers; for example, the query "What is the best way to remove stains?" may lead to a hallucinated answer from a user's point of view if the LLM replies "you can remove stains from clothes with product xyz" whereas the expected answer was to do so from the fabric of a sofa. Additionally, we posit that an LLM receiving an **unseen instruction type** (an instruction not seen at training time) can also induce a hallucination simply by not knowing what action or task to carry out, as in the prompt: "let's play Werewolf" which is a game that is not frequently used for training LLMs.

A further cause of hallucinations is the phenomenon of **knowledge overshadow** (Zhang et al., 2025b) occurring when prompts contain more than one entity: The highest-frequency entity in pretraining data becomes the *dominant* entity, overshadowing the less frequent ones, which can result in a

hallucination if the main entity in the query is not the dominant one. As pinpointed in (Zhang et al., 2025b), given a query such as "Between the Arc de Triomphe and the Musée d'Orsay, which landmark is more popular in France?" if France overshadows the Arc de Triomphe and the Musée d'Orsay, then the LLM may interpret the query as requesting information about a popular landmark in France for which the LLM may output one of the most popular such as the Louvre or the Eiffel Tower. The last prompt feature we consider can actually have a positive impact on LLMs: Adding **in-context examples** to guide an LLM's generation by showing both the principles of the task and the expected response can reduce the chance of hallucination (Pal et al., 2023). For instance, if a prompt contains an unseen instruction such as "write a poem about the Earth" but the in-context examples demonstrate the expected output as in "write a poem about the Sun. The Sun is a beautiful star which warms us at day" then the LLM will use these examples to better contextualize the query by extracting the intent, inferring the input-output process, and mimicking the output.

Ultimately, the last salient cause of hallucinations from inference time we address is the problem of **needle-in-a-haystack**: Finding a piece of key information that can serve as an answer to a question – a *needle* – within a wide body of information in the context – a *haystack* (Banerjee et al., 2024). It has been shown that the accuracy of retrieving a needle declines a) with increased context length, b) when the needle is placed at the beginning or middle of the haystack, and c) with smaller needle lengths (Bianchi et al., 2025; Schuster et al., 2025). Thus, depending on these attributes is the difficulty of retrieving the target piece of information from a lengthy input context and failure to do so translates into a hallucination.

5 Conclusions

We presented a theoretical model of hallucinations in NLP LLMs: An abstract structure with its underlying rationale of how a set of interconnected factors gives rise to this phenomenon. A theoretical model, while well grounded on empirical findings, needs not be purely empirically driven – hence its name *theoretical* – as it will also abstract the domain knowledge of researchers whom, on intellectual grounds, theorize about relationships and factors from which new findings can be derived via

future experiments. We claim, the NLP field has studied phenomena mainly experimentally, leaving a plethora of scattered discoveries; synthesizing these works into conceptual representations, as done here, can provide us with a better conceptual understanding of how each phenomenon emerges, thus advancing our field.

Limitations

One natural limitation of this work is the possibility of not being exhaustively comprehensive and thus not including one or more causes of hallucination in NLP LLMs. As new research is published, new causes emerge. Nevertheless, the proposed theoretical model can be easily updated and modified by other researchers in the field satisfying any field-related idiosyncrasies from where the new research has been published. Thereof, we leave as an open, future work the continuous update of this model by all interested researchers.

The theoretical model may embody hypothetical relations or factors which may be proven inaccurate in future works. This is precisely one objective of this model. Any theorizing is not randomly proposed; it comes from our domain knowledge; it is to the best of our intellectual effort that we consider such hypotheses to hold true, aligned to the evidence from previous works, in the hope that future work will prove them correct or incorrect, as in the way any scientific field advances.

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A Appendix

A.1 Further Assumptions and Boundaries

We further delimit the scope under which our theoretical model is valid. We do not consider outer indirect factors such as the manufacturer of the LLM which may influence its programming quality. Furthermore, we view an LLM as a standalone system, and so we do not consider external sources or methods such as Retrieval Augmented Generation (RAG). Additionally, since our model only concerns with explanation of hallucinations, we do not consider any mitigation method. In addition, we define an input prompt as the composition: prompt = in-context learning examples + context + query, where 1) the operator ‘+’ means text concatenation, 2) a query can also be an instruction, and 3) the only element that must always appear in a prompt is the query (the other elements are optional). In this paper, if no specific element is specified in the example prompts provided, then it is assumed that the prompt only contains a query or an instruction.

A.2 Tables

Origin	Factor	Example or Scenario
Pretraining data	Multilingualism	As mentioned in Section 4.1.1, depending on the amount of multilingual tokens in the data is the magnitude of the effect of multilingualism on causing hallucinations such as input queries with complex English syntactic structures eliciting hallucinations due to the LLM’s inability of correctly parsing them caused by the impairment mentioned in Section 4.1.1.
Pretraining	Loss function	As mentioned in Section 4.1.2, the loss function does not incentivize factuality; thus, even when pretraining using only factual data, hallucinations can still occur when the LLM is queried with novel concepts or entities because this leads the LLM to infer new information from its parametric memory, and to do so, the LLM will simply repeat the learned generation process of predicting tokens based on patterns recovered from data.
Pretraining	Not learning the pretraining data distribution	As mentioned in Section 4.1.2, one possible consequence of not accurately learning the pretraining data distribution is the LLMs’ inability to faithfully capture grammar rules or tenses; thus, with a query in present perfect tense such as "Who has been the president of the USA for the last 3 years?" An LLM may interpret this as a past tense structure, and answer it with the name of the president 3 years ago which may have been another person if in that year there was a change of administration.
SFT data	Amount of SFT data	As mentioned in Section 4.2.1, even for simple tasks, insufficient data can lead to hallucinations; for example, given the prompt: "Translate the following question to French: what is the capital of France?" the LLM may misinterpret it as a question seeking for a capital city and respond "Paris" which is the correct answer to the question "what is the capital of France?" picked from pretraining, but it is an incorrect answer to the target instruction requesting a translation.

Table 1: Examples of hallucinations, or cases when hallucinations may arise, for selected factors.

Origin	Factor	Example or Scenario
SFT	Not learning the SFT data distribution	As mentioned in Section 4.2.2, not accurately picking the right statistical cues and patterns due to overfitting can lead to hallucinations; for example, given the instruction "Write a list of the top 40 most famous songs of 2026" an LLM may return a list of only 10 songs if it overfitted to the SFT examples requesting for lists of 10 items across diverse domains such as the 10 best movies or the 10 most popular cars.
LLM architecture	Reasoning capabilities	As mentioned in Section 4.4, failure in the capability to carry out a complex reasoning for complex queries, such as some historiographical queries, can lead to providing an incorrect answer and thus a hallucination; for example, for the query "What were the motivations of those whose actions contributed to the onset of the First World War?" an LLM needs to first figure out the actions that caused the first world war, then find whom were behind those actions, and finally discover their motivations.
LLM architecture	Vocabulary size	As mentioned in Section 4.4, a smaller and coarser vocabulary can lead to some scenarios of hallucination, as for example, with the query "Is magpie the favorite among people?" since 'magpie', the bird, may be tokenized as 'mag' (short for magazine) and 'pie' (the dessert) with no concise way to disambiguate due to the unclear context; but, this ambiguity may not happen if the vocabulary included 'magpie'.
Inference time	Prediction consistency	As pinpointed in (Zhang et al., 2024), given an input query such as "is 2^{10} greater than 10^3 ?" LLMs may incorrectly respond "no" and then give a reason such as "because the base of 10 grows faster than the base of 2", continuing the hallucination despite knowing the correct computations, which can be verified by querying the LLMs with "what is 2^{10} ?", "what is 10^3 ?", and then "is 1024 greater than 1000?" where LLMs respond "yes".

Table 2: Continuation of examples of hallucinations, or cases when hallucinations may arise, for selected factors.

Factors		Relation	Proposition
Antecedent	Consequent		
Data quality	Imitative falsehoods	causal	The presence of inaccurate facts, biases, noisy data in the pretraining corpus, i.e. <i>falsehoods</i> , will cause LLMs to incorporate the ability to imitate such falsehoods into their behavioral repertoire
Data quality	Parameters (updates)	embodiment	The quality of features and contexts info from entities and concepts, which will be learned and encoded in the parameters, depend on the data quality
Subjective information	Data quality	influence	For some domains, subjective beliefs from authors may take the form of biases or inaccurate facts which can in turn contribute to the pool of falsehoods
Amount of pretraining data	Parameters (updates)	causal	The number of pretraining documents will affect the quality of the knowledge in the parameters (e.g. the less the documents, the less the contextual information coverage)
Domain of pretraining data	Data quality	influence	The domain of the pretraining corpus may influence the likelihood of observing biases, inaccurate facts, certain writing style, vagueness, ambiguity, or noise in the corpus
Domain of pretraining data	Subjective information	influence	The domain of the pretraining corpus may influence the objectivity of the data (e.g. scientific domains may be more objective than social media domains)
Domain of pretraining data	Amount of pretraining data	influence	The domain of the pretraining corpus may be related to the amount of available data (e.g. social media documents are more abundant than scientific documents)
Long-tail entities	Parameters (updates)	embodiment	Highly infrequent entities will result in their embeddings having poor quality (not storing enough contextual information about such entities)
Outdated data	Parameters (updates)	causal	An outdated pretraining corpus will cause the embeddings of the entities and concepts to contain outdated information
Multilingual data	Parameters (updates)	causal	Variations in the linguistic patterns of other languages in the pretraining corpus may conflict with English patterns causing conflicts in the encoded information in the parameters

Table 3: Edges in our theoretical model of hallucinations (pretraining stage) and their corresponding relation type and propositional statement.

Factors		Relation	Proposition
Antecedent	Consequent		
Incomplete information	Parameters (updates)	causal	Missing information of entities or concepts in the pretraining corpus will cause a poor or incomplete contextualization of their embeddings
Statistical co-occurrences	Parameters (updates)	embodiment	The more frequently two or more entities co-occur in the pretraining corpus, the higher the chance of capturing this statistical association in the parameters
Number of pretraining epochs	Parameters (updates)	causal	Under some conditions, a sufficient number of pretraining epochs will cause the parameters to optimally or accurately encode contextual information
Imitative falsehoods	Parameters (updates)	embodiment	The expression of the behavior of imitative falsehoods will be embodied by the LLM via the update of parameters after pretraining
Teacher forcing	Parameters (updates)	embodiment	The forced <i>expectation</i> of the LLM to receive always a golden input sequence to predict the next token will be embodied in the parameters
Loss function	Parameters (updates)	causal	The autoregressive loss function causes the prediction of tokens to be based on statistical/linguistic patterns which leads the parameters to a certain space to accurately learn those patterns
Parameters (updates)	Distribution not learned	mediation	Antecedent factors embodied in the parameters may cause or influence the LLM to not properly learn the distribution of the pretraining data, and is via the current configuration of the parameters that this takes effect
Parameters (updates)	Hallucination	mediation	The parameters are one way through which the previous antecedents can take effect. After pretraining, the parameters will contain the combined effects of all the antecedent causes of hallucination which will be triggered after the LLM receives an input prompt
Distribution not learned	Hallucination	causal	If the distribution of the pretraining data is not properly learned, it will lead the LLM to not properly recovering the linguistic or statistical patterns required to generate a factual or faithful prediction

Table 4: Continuation of edges in our theoretical model of hallucinations (pretraining stage) and their corresponding relation type and propositional statement.

Factors			Relation	Proposition
Antecedent		Consequent		
Incorrect labels		Parameters (updates)	causal	The pairing of inputs with incorrect labels, derived from a wrong annotation scheme or human error, will be learned and encoded in the parameters
Amount of SFT data		Parameters (updates)	causal	Insufficient instances to learn a specific domain or task may lead to a sub-optimal parameter configuration
New knowledge		Parameters (updates)	causal	New knowledge in the SFT data (not seen during pretraining) may either shift, modify, or significantly alter the pretraining distribution encoded in the parameters
Instruction types		Parameters (updates)	causal	If the SFT data contains instructions that are significantly complex, and/or the diversity of instruction types is high, then these instances may lead the training to a sub-optimal parameter space causing the tasks represented in these instances to not be properly learned
Creative data (SFT)	Innate	diverge	causal	Instances which show the LLM to behave in <i>creative</i> ways by purposefully diverging from accurate or factual information will lead to the phenomenon of innate divergence
Innate divergence		Parameters (updates)	embodiment	The repeated exposure to <i>creative</i> instances, such as open-ended dialogues, can lead to embodying the behavior of purposefully generating creative sequences, possibly diverging from factuality

Table 5: Edges in our theoretical model of hallucinations (SFT stage) and their corresponding relation type and propositional statement.

Factors		Relation	Proposition
Antecedent	Consequent		
Number of fine-tuning epochs	Parameters (updates)	causal	Insufficient training epochs can lead to underfitting while the opposite can lead to overfitting; any scenario will shift the parameters to a specific parameter space reflecting characteristics of such a scenario such as learning noise from the data
Parameters (updates)	Distribution not learned	mediation	Similar to the pretraining scenario, particular states of antecedent factors of the parameters may lead to not properly or accurately recovering all the statistics and patterns from the SFT data, leading to an imprecise embodiment of the SFT data distribution
Parameters (updates)	Catastrophic forgetting	causal	Learning new knowledge at SFT time can lead to a parameter configuration which causes previously learned abilities or knowledge to be destroyed, overridden, or forgotten
Parameters (updates)	Hallucination	mediation	The encoded effects of the antecedents of the parameters will indirectly and effectively lead LLMs to hallucinate when an input prompt meets the conditions for triggering one or several of them
Catastrophic forgetting	Hallucination	causal	Significant alteration or forgetting of previous knowledge from pretraining can lead to hallucinations
Distribution not learned	Hallucination	causal	Incorrectly or poorly modeled statistical, linguistic, and semantic patterns from the data will lead to hallucinations when the concepts or tasks involved in such patterns are invoked via an input prompt

Table 6: Continuation of edges in our theoretical model of hallucinations (SFT stage) and their corresponding relation type and propositional statement.

Factors		Relation	Proposition
Antecedent	Consequent		
Incorrect annotation or reward	Misleading alignment	causal	Systematically skewed annotations which favor a specific response type may lead to an underlying model of misleading alignment by showing the LLM to align to aspects such as fluency or politeness that may take over factuality
Incorrect annotation or reward	Reward hacking	causal	Systematically skewed rewards which favor a specific response type may lead to an underlying model of reward hacking by incentivizing the LLM to align to aspects such as fluency or politeness that may take over factuality
Amount of alignment data	Parameters (updates)	causal	The number of instances will be partly responsible for determining the optimality in the final configuration of the parameters in the parameter space
Creative data (human alignment)	Sycophancy	causal	Instances which favor some type of creative outputs, such as responses in the form of funny dialogues, may promote the sycophantic behavior of LLMs by rewarding the LLMs to generate outputs that may jeopardize factuality
Misleading alignment	Sycophancy	causal	The underlying misleading alignment in the data to favor human preferences or values over factuality can result in the phenomenon of sycophancy
Reward hacking	Sycophancy	causal	The underlying alignment in the skewed reward function to favor human preferences or values over factuality can result in the phenomenon of sycophancy

Table 7: Edges in our theoretical model of hallucinations (alignment to human preferences stage) and their corresponding relation type and propositional statement.

Factors		Relation	Proposition
Antecedent	Consequent		
Sycophancy	Parameters (updates)	embodiment	The behavior of sycophancy will be embodied in the parameters which will be activated with any input prompt that asks to generate an output that follows the human preferences learned from alignment training
Number of alignment epochs	Parameters (updates)	causal	The number of epochs is a direct guide to the parameter configuration in the parameter landscape
Number of alignment epochs	Mode collapse	causal	The more the number of epochs for training, the more certain it is the occurrence of the mode collapse phenomenon due to overfitting
Mode collapse	Parameters (updates)	embodiment	The behavior associated with the phenomenon of mode collapse will be embodied in the parameters which will be invoked when an input prompt refers to or contains any element or pattern overfitted during training
Parameters (updates)	Distribution not learned	mediation	Similar to the pretraining and SFT scenarios, the antecedent factors of the parameters may lead to not properly or not accurately recovering all the statistical cues as well as the linguistic and semantic patterns from the alignment data; this in turn, will lead to not properly learning and embodying the data distribution
Parameters (updates)	Hallucination	mediation	The embodiment of all the antecedents of the parameters can indirectly and effectively lead LLMs to hallucinate when an input prompt meets the conditions for triggering one or several of them
Distribution not learned	Hallucination	causal	Incorrectly or poorly learned statistical, linguistic, and semantic patterns can lead to hallucinations when the concepts or tasks involved in such patterns are elicited via an input prompt

Table 8: Continuation of edges in our theoretical model of hallucinations (alignment to human preferences stage) and their corresponding relation type and propositional statement.

Factors		Relation	Proposition
Antecedent	Consequent		
Transformer model	Positional bias	causal	Features from the Transformer model such as positional encoding, attention model, mask type, computation of K, V matrices, or use of softmax for normalization may impose a certain type of positional bias
Transformer model	Long-range dependencies	causal	Factors such as attention dilution, weakened gradient signals, softmax attention decay, context fragmentation, or positional encoding can contribute to the occurrence of long-range dependencies
Positional bias	Recency bias	instantiation	Recency bias is a type of positional bias where most recent tokens or tokens near the end of an input sequence receive more attention
Positional bias	Short-range bias	instantiation	Short-range is a type of positional bias where tokens in a localized cluster inside an input sequence receive bigger attention weights due to their proximal position
Positional bias	Attention sink	instantiation	Attention sink is a type of positional bias where tokens at the beginning of an input sequence receive bigger attention weights
Positional bias	Lost-in-the-middle	instantiation	Lost-in-the-middle is a type of positional bias where tokens in the middle of an input sequence receive a significantly smaller attention weight than those at the beginning or end
Recency bias	Lost-in-the-middle	causal	Recency bias is a plausible cause of the phenomenon of lost-in-the-middle by directing the attention towards tokens at the end of an input sequence
Attention sink	Lost-in-the-middle	causal	Attention sink is a plausible cause of the phenomenon of lost-in-the-middle by directing the attention towards tokens at the beginning of an input sequence
Recency bias	Hallucination	causal	Recency bias may cause hallucinations if the key information to be attended to in order to generate a factual or faithful answer is placed at the begging of an input sequence
Short-range bias	Hallucination	causal	Short-range bias may cause hallucinations if the pieces of information to be attended to, in order to generate a factual or faithful answer, are placed distantly enough to be in different localized clusters of tokens

Table 9: Edges in our theoretical model of hallucinations (due to the architecture of LLMs) and their corresponding relation type and propositional statement.

Factors		Relation	Proposition
Antecedent	Consequent		
Domain of pretraining data	Long-range dependencies	causal	Documents from domains such as web content may lead to the long-range dependencies problem since they do not systematically contain semantically-valid long-range dependencies due to their simplified or inconsistent writing style or due to being spuriously concatenated to yield larger documents
Attention sink	Hallucination	causal	Attention sink may cause hallucinations if the key information to be attended to, in order to generate a factual or faithful answer, is placed towards the end of the input sequence
Lost-in-the-middle	Hallucination	causal	Lost-in-the-middle may cause hallucinations if the key information to be attended to in order to generate a factual or faithful answer is placed at the middle of the input sequence
Transformer model	Insufficient context attention	causal	Features from the Transformer model such as positional encoding, attention model, mask type, or computation of K, V matrices can impose a restriction to pay more attention to the parametric memory than to the input context
Insufficient context attention	Hallucination	causal	When the LLM pays more attention to its parametric knowledge than to the input context it may induce unfaithful hallucinations if this knowledge contradicts or deviates from the target information of the input context to be extracted as an answer
Reasoning capabilities	Hallucination	causal	Limitations in terms of reasoning ability directly impact on hallucinations as the complexity of the input query increases
Imperfect encoding	Hallucination	causal	Due to intrinsic limitations in the mechanism for building the embedding representations of entities and concepts, the encoding may not derive in a perfect <i>memory</i> of an entity or a concept which may cause hallucinations at inference time when the LLM is asked about an imperfectly encoded attribute
LLM size	Hallucination	causal	The number of learnable parameters in an LLM has an impact on the generation of hallucinations due to the amount of factual information that can be stored and the capacity for faithfully capturing the data distribution
Vocabulary size	Hallucination	causal	The level of granularity of the vocabulary may impact on the occurrence of hallucinations generated; the finer-grained the vocabulary the less hallucinations

Table 10: Continuation of edges in our theoretical model of hallucinations (due to the architecture of LLMs) and their corresponding relation type and propositional statement.

Factors		Relation	Proposition
Antecedent	Consequent		
Softmax bottleneck	Stochastic decoding	influence	Some of the candidate tokens with the highest likelihood or probability to be chosen as a prediction may not be a faithful choice and depending on the decoding method it is the likelihood for generating a hallucination
Stochastic decoding	Hallucination	mediation	Any stochastic decoding method may induce hallucinations, where sampling-induced randomness acts as a mediator due to selecting tokens within the high-probability set constrained by the softmax bottleneck
Prediction consistency	Hallucination	causal	In a sequence generation, the factuality of the first tokens may determine the factuality of the remaining sequence generated, and if the LLM is consistent with its first tokens predicted and these tokens are a hallucination then the remaining predictions will continue the hallucination
Short-range bias	Prediction consistency	influence	The prediction path set by the first predicted tokens may be influenced by or reinforced due to the effect of short-range bias
Over-confidence	Prediction consistency	influence	The prediction of tokens with overconfidence may influence the LLM to keep in the thread started from the first predicted tokens
Teacher forcing	Exposure bias	causal	The mechanism of <i>feeding</i> the LLM with a golden input sequence to predict the next token is precisely what gives rise to the exposure bias issue due to the difference between receiving such golden input with respect to receiving the previously generated sequence as input at inference time

Table 11: Edges in our theoretical model of hallucinations (arising at inference time) and their corresponding relation type and propositional statement.

Factors		Relation	Proposition
Antecedent	Consequent		
Exposure bias	Over-confidence	influence	The phenomenon of exposure bias may influence the overconfidence of LLMs predictions by incrementing the confidence of the current prediction when given as input its own previous predictions matching an already known, familiar, or similar sequence
Exposure bias	Prediction consistency	influence	The fact that LLMs are trained to predict based on a gold input sequence but at inference time they predict based on their own generations may influence the prediction consistency by not being able to control and self-correct the current generation
Knowledge overshadow	Hallucination	causal	The presence of more than one entity in an input query can induce hallucinations if the query's most frequent entity in pretraining data is not the main entity in the query, thus becoming a dominant entity overshadowing the main entity due to a possible <i>misdirection</i> of the LLM focus
Needle-in-a-haystack	Hallucination	causal	The problem of finding specific key information within a long input context can cause a hallucination depending on the complexity of the problem. Varying factors such as context length or position of key information can impact on the likelihood of inducing the hallucination
Prompt features	Hallucination	causal	Several features such as the specificity and precision of the query and context, whether the instruction was unseen at training time, the input query length, among others, can trigger hallucinations. Notably, in-context examples can reduce the likelihood of hallucination by providing key cues to better ground the expected answer

Table 12: Continuation of edges in our theoretical model of hallucinations (arising at inference time) and their corresponding relation type and propositional statement.

Factors		Relation	Proposition
Antecedent	Consequent		
Number of pretraining epochs	Amount of pretraining data	interaction	The strength of the effect of the amount of data on hallucinations depends on the number of epochs: While huge amounts of data are beneficial, to some extent, for LLMs to guard against hallucinations, it depends on the right number of pretraining cycles to ensure an optimal learning of the information and rich contextualization of embeddings
Amount of pretraining data	Multilingual data	interaction	Depending on the amount of data in languages with significant variations from English is the magnitude of the effect on altering English patterns and thus causing hallucinations
Number of pretraining epochs	Multilingual data	interaction	Depending on how many epochs the parameters are trained using data in languages with significant variations from English is the likelihood of the effect on altering English patterns and thus causing hallucinations
Loss function	Teacher forcing	interaction	Depending on the loss function used is the possibility of using teacher forcing for pretraining, i.e. with cross-entropy is possible to apply the teacher-forcing mechanism for pretraining
LLM size	Imitative falsehoods	interaction	The accuracy in modeling an imitative behavior of falsehoods from pretraining data may depend on the number of parameters of the LLM: Bigger LLMs may better learn to model such falsehoods than smaller LLMs
LLM size	Data quality	interaction	Depending on the size of the LLM is the effect of the quality of the data on the hallucinations: Optimal data quality with large or small LLMs will lead to an optimal scenario reducing hallucinations, but deficient data quality will have a bigger impact on hallucinations with large LLMs since they can better capture noise and biases than smaller LLMs
Number of fine-tuning epochs	Amount of SFT data	interaction	The appropriate fine-tuning to avoid hallucinations depends on the right number of epochs and right amount of data
Incorrect labels	Amount of SFT data	interaction	The more the number of instances with incorrect labels, the higher the change of inducing hallucinations due to more exposure to incorrect input-output mappings at training time

Table 13: Edges in our theoretical model of hallucinations (interactions between factors) and their corresponding relation type and propositional statement.

Factors		Relation		Proposition
Antecedent	Consequent			
New knowledge	Amount of SFT data	of	interaction	The more the number of instances containing new knowledge the higher the change of inducing hallucinations due to a bigger likelihood of destroying or altering learned patterns from pretraining or inducing catastrophic forgetting
Instruction types	Amount of SFT data	of	interaction	The more the number of different types of instances from different tasks or domains the higher the risk of inducing hallucinations due to requiring a bigger amount of gradient signal for each task type to accurately learn them
Creative data (SFT)	Amount of SFT data	of	interaction	The more the number of instances containing creative content that diverges from factuality, such as open-ended dialogues, the higher the chance of inducing hallucinations due to promoting diverging from factuality
Number of alignment epochs	Amount of alignment data	of	interaction	Dependent on the number of epochs and amount of alignment data is the correct modeling of the alignment to human preferences and thus the impact on generating hallucinations (we note this interaction further depends on the alignment method used such as DPO, RLHF, or other)
Creative data (human alignment)	Amount of alignment data	of	interaction	The more the number of instances containing creative content, the higher the chance of aligning to factors such as creativity or politeness at the risk of overlooking factuality in order to please humans
Incorrect annotation or reward	Amount of alignment data	of	interaction	The more the number of incorrectly annotated instances or incorrectly modeled rewards, the higher the chance of emerging misleading alignments or reward hacking, and thus the higher the risk of hallucinating
Recency bias	Attention sink		interaction	The synergistic interaction of these two factors can lead to the phenomenon of lost-in-the-middle
Positional bias	Domain data	of	interaction	Features from the pretraining documents such as writing style, among others, can further increase the effect of positional bias by, for example, pre-training with documents in which the first sentence always summarize each document, or where the most important key information is placed at the middle of the documents

Table 14: Continuation of edges in our theoretical model of hallucinations (interactions between factors) and their corresponding relation type and propositional statement.

Factors		Relation	Proposition
Antecedent	Consequent		
Long-range dependencies	Short-range bias	interaction	The localized effect from the short-range bias may aggravate the small attention weight between two far away but semantically related tokens happening due to their distance, which further increases the chance of hallucination
Vocabulary size	Amount of pretraining data	interaction	Depending on the amount of data is the quality of each token (embedding) in the vocabulary (a large vocabulary with a small corpus will lead to poorly contextualized representations) and thus it may influence the choice of the vocabulary size
Vocabulary size	Number of pretraining epochs	interaction	The larger the vocabulary size, the more the number of epochs that may be required to accurately encode information in the embeddings of entities and concepts
LLM size	Knowledge overshadow	interaction	The larger the LLM the more the tendency of dominant entities to overshadow infrequent entities in input prompts due to the diminishing ability of the LLMs to retain semantical knowledge of unpopular entities, leading to more hallucinations
Needle-in-a-haystack	Prompt features	interaction	Features from input prompts such as the length of the context, or the query type, can increase the difficulty of finding the key information required to answer the input query, leading to a bigger likelihood of hallucination
Needle-in-a-haystack	Positional bias	interaction	The difficulty of retrieving key information from the input context not only depends on its position in the context, but it also depends on the most prominent type of positional bias embodied by an LLM, which can further aggravate the difficulty of retrieving this information leading to a higher chance of hallucination
Needle-in-a-haystack	Long-range dependencies	interaction	If the key information to be retrieved encompasses two subsequences which are separated by a long distance inside the input context, then due to the long-range dependencies problem the likelihood of recovering the complete information decreases and as such the chance of hallucination increases

Table 15: Continuation of edges in our theoretical model of hallucinations (interactions between factors) and their corresponding relation type and propositional statement.